

WORKED KEY

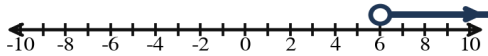
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Period: _____

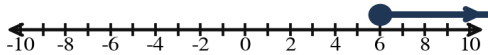
This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Which graph shows the solution set of $x \leq 6$?

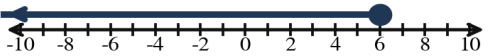
(A)



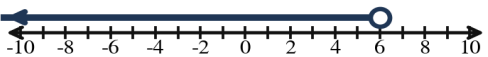
(B)



(C)



(D)



Answer: $x \leq 6$: CLOSED circle at 6 (included), shaded to the LEFT

2. Solve and graph: $x + 8 < 2$

Answer: $x < -6$ — open circle at -6 , shade left

3. Solve and graph: $6y - 1 < 7y + 6$

Answer: $-1 < y + 6 \rightarrow -7 < y$, so $y > -7$ — open circle at -7 , shade right

4. Solve: $-3x + 7 \leq -14$

Answer: $-3x \leq -21 \rightarrow$ divide by -3 and FLIP: $x \geq 7$

5. Solve: $6(x - 4) > -18$

Answer: $6x - 24 > -18 \rightarrow 6x > 6 \rightarrow x > 1$

6. Solve: $7x - 6 \leq 3x + 2$

Answer: $4x \leq 8 \rightarrow x \leq 2$

7. Solve the compound inequality: $-7 < x + 2 \leq -3$

Answer: subtract 2 from all parts: $-9 < x \leq -5$

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8. Solve the compound inequality: $x + 6 \leq 4$ or $x - 3 \geq 5$

Answer: $x \leq -2$ or $x \geq 8$

9. A rectangular banner has width 7 ft, and the ribbon border sewn around its edge can be at most 40 ft long. Write and solve an inequality to find the possible lengths.

Answer: $2l + 14 \leq 40 \rightarrow 2l \leq 26 \rightarrow l \leq 13$ ft

10. Tomás has \$18 saved and earns \$14 each week walking dogs. He needs AT LEAST \$116 for a guitar amplifier. Write and solve an inequality to find how many weeks he must save.

Answer: $14w + 18 \geq 116 \rightarrow 14w \geq 98 \rightarrow w \geq 7$ weeks

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11. Error Analysis — Rex graphed $x < 5$ with a CLOSED circle at 5 and shading to the RIGHT. Identify the mistake.

Answer: two mistakes: $<$ EXCLUDES 5, so the circle must be OPEN, and 'less than' shades to the LEFT

12. A bakery's dough proofing box must stay between 78°F and 86°F , inclusive. Write a compound inequality for the allowed temperatures t .

Answer: $78 \leq t \leq 86$

13. Solve the compound inequality: $2 \leq 5x - 8 < 27$

Answer: add 8 to all parts: $10 \leq 5x < 35 \rightarrow 2 \leq x < 7$

Cumulative Review — ACT Style. Circle the best answer.

14. Which is the equation of the line with slope 5 and y-intercept -1 ?

(A) $y = -x + 5$

(B) $y = 5x + 1$

(C) **$y = 5x - 1$**

(D) $y = x - 5$

Answer: $m = 5$ goes with x ; $b = -1$ stands alone

15. What is the slope of the line through $(-2, 5)$ and $(4, -7)$?

(E) **-2**

(F) 2

(G) $-1/2$

(H) 6

Answer: $(-7 - 5)/(4 - (-2)) = -12/6 = -2$

16. Let $f(x) = 8x - 5$. What is $f(4)$?

(A) 7

(B) **27**

(C) 37

(D) -5

Answer: $32 - 5 = 27$